

MADHUSMITA TALUKDAR

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SUMMARY

Highly analytical and detail-oriented aspiring Data Scientist with strong foundations in machine learning, predictive analytics, and data-driven problem solving. Experienced in building end-to-end ML solutions, optimizing data pipelines, and developing models for risk detection, fraud analysis, and large-scale decision support. Adept at transforming raw data into actionable insights and designing scalable, reliable systems aligned with real-world business needs. Strong focus on model accuracy, performance, and deployability, with a clear understanding of how data science supports secure, high-impact applications.

EDUCATION

LOVELY PROFESSIONAL UNIVERSITY

Bachelor of Technology (B.Tech.) in Computer Science Engineering

Punjab, India

Expected August 2027

PROJECTS

Smart URL Safety Checker | Python · Flask · Scikit-learn · XGBoost · Pandas · NumPy · REST API

Github Link

Jun' – Jul' 2025

- Developed a machine learning-based web application to detect and classify malicious URLs for phishing prevention.
- **Designed** a feature extraction pipeline generating 30+ lexical, host-based and other attributes from 70K+ URLs using Python & Pandas for large scale securing data.
- **Trained** and evaluated multiple ML models (Random Forest, XGBoost, SVM, Logistic Regression, KNN) achieving up to 97% accuracy.
- **Deployed** the risk model as a scalable REST API endpoint using Flask, demonstrating end-to-end MLOps capability.

What2Wear AI Chatbot | Python · JavaScript · HTML/CSS · API Integration

Github Link

May' 2025

- **Implemented** a rule-based recommendation engine that personalized outfit suggestions by integrating external weather API data and user preferences.
- **Optimized** the decision logic, improving recommendation relevance to **92%** and enhancing the overall user experience.
- **Ensured** high performance by achieving a **30%** faster response and inference speed, crucial for low-latency user interaction.
- Developed a clean, responsive user interface with HTML, CSS, and JavaScript to ensure smooth and intuitive interaction with the chatbot.

Analyzing Crime Data for City Safety | Python · Pandas · NumPy · Scikit-learn · Matplotlib · Seaborn ·

GeoPandas · Google Colab

Github Link

Apr' – May' 2025

- **Constructed** a data-driven analytical model to uncover complex temporal and spatial crime patterns for strategic decision support.
- **Managed** the preprocessing and cleaning of **250K+** geographical crime records, reducing data noise by **35%** through rigorous feature engineering.
- **Applied** clustering and trend-analysis techniques, achieving a Silhouette Score of 0.92 for accurate hotspot identification.
- **Visualized** high-impact findings using Matplotlib and Seaborn, translating complex data into actionable public safety insights.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, C/C++, Java
- **ML/DS Frameworks:** Scikit-learn, TensorFlow, XGBoost, Pandas, NumPy, Matplotlib, Seaborn
- **Databases & Backend:** MySQL, PostgreSQL, Flask, Data Pipeline
- **Web & Application Development:** HTML5, CSS3, JavaScript
- **Tools & Platforms:** Jupyter Notebook, Google Colab, Power BI, Excel, Git/GitHub, VS Code
- **Core Concepts:** Data Structures & Algorithms (DSA), Object Oriented Programming (OOP), **MLOps Fundamentals**, Data Structures & Algorithms (DSA), **Predictive Modeling, Feature Engineering, Model Optimization**, Cybersecurity Fundamentals, **Risk Analysis**, Operating Systems, Computer Network, AI Foundation

CERTIFICATIONS & TRAINING

- **SQL(intermediate) certificate - HackerRank** [Link to Certificate](#)
- **Human Computer Interaction- NPTEL** [Link to Certificate](#)
- **The Bits and Bites of Computer Networking -Coursera** [Link to Certificate](#)

AWARDS & ACHIEVEMENTS

- **Authored** a research paper on the security implications of integrating **Windows Copilot** into the operating system
- **Worked** as a **core member** and **graphic designer** in **Cisco iGen community** and **Encrypt Edge**.

STRENGTHS

- Analytical thinking and structured problem-solving
- Fast Learner with Adaptability
- Creative Problem Solver